

Git and GitHub Commands

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What is Git?

Git is a version control system that tracks changes in code and manages project history.

What is GitHub?

GitHub is a cloud-based platform for hosting Git repositories, enabling collaboration and sharing.

Git and GitHub together make coding projects organized, trackable, and collaborative.

- Use Git to manage versions of your code locally.
- Use GitHub to share and collaborate with others online.
- Regularly check with commands like `git status` and `git log` to stay aware of changes.
- Push and pull often to keep your work synchronized with the team.
- Branching and merging allow safe experimentation without disturbing the main project.

Git Commands :

- `git init` :Initializes a new Git repository. Example: `git init`
- `git status` :Shows the current state of the repository. Example: `git status`
- `git add .` : Stages all changes for commit. Example: `git add .`

- `git commit -m`: Saves changes with a message. Example: `git commit -m "Initial commit"`
- `git branch` :Lists or creates branches. Example: `git branch new-feature`
- `git branch -M main`: Renames branch to main. Example: `git branch -M main`
- `git remote add origin` :Links local repo to remote. Example: `git remote add origin https://github.com/user/repo.git`
- `git remote -v` Shows remote connections. Example: `git remote -v`
- `git push -u origin main` Pushes code to remote main branch. Example: `git push -u origin main`
- `git pull origin main`: Fetches and merges changes from remote. Example: `git pull origin main`
- `git config --list` Shows all Git configurations. Example: `git config --list`
- `git config user.name` :Sets or shows username. Example: `git config user.name "Mrunmayee Shimpi"`
- `git config user.email` :Sets or shows email. Example: `git config user.email mrunmayee1605@example.com`
- `git clone` :Copies a repository from remote. Example: `git clone https://github.com/user/repo.git`